

1. (30 points) Find the inverse of $A = \begin{bmatrix} 3 & 0 & -2 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{bmatrix}$ if it exists.

$$|A| = 3 \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} + (-2) \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 0 + 0 + (-2)(2) = -4 \neq 0$$

↓
invertible

Now, we find the determinants of each 2×2 matrices

for the elements of first row:

$$a_{11} = \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = 0 \quad a_{12} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2 \quad a_{13} = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2$$

Second row:

$$a_{21} = \begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix} = 2 \quad a_{22} = \begin{vmatrix} 3 & -2 \\ -1 & 1 \end{vmatrix} = 1 \quad a_{23} = \begin{vmatrix} 3 & 0 \\ -1 & 1 \end{vmatrix} = 3$$

Third row:

$$a_{31} = \begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix} = 2 \quad a_{32} = \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} = 5 \quad a_{33} = \begin{vmatrix} 3 & 0 \\ 1 & 1 \end{vmatrix} = 3$$

$$\text{Adj}(A) = \begin{bmatrix} 0 & -2 & 2 \\ -2 & 1 & -3 \\ 2 & -5 & 3 \end{bmatrix}^T$$

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A) = -\frac{1}{4} \begin{bmatrix} 0 & -2 & 2 \\ -2 & 1 & -3 \\ 2 & -3 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} & \frac{3}{4} \\ -\frac{1}{2} & \frac{3}{4} & -\frac{3}{4} \end{bmatrix}$$

A is invertible? $|A| \neq 0$

$|A| = -4 \neq 0 \rightarrow A$ is invertible

$$\left[\begin{array}{ccc|ccc} 3 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ -1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\frac{1}{3}R_1 \rightarrow R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ -1 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{R_2 - R_1 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & \frac{5}{3} & -\frac{1}{3} & 1 & 0 \\ -1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & \frac{5}{3} & -\frac{1}{3} & 1 & 0 \\ 0 & 1 & \frac{1}{3} & \frac{1}{3} & 0 & 1 \end{array} \right]$$

$$\xrightarrow{R_3 - R_2 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 1 & \frac{5}{3} & -\frac{1}{3} & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{3}{4} & -\frac{3}{4} \end{array} \right] \xrightarrow{R_1 + \frac{2}{3}R_3 \rightarrow R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & \frac{5}{3} & -\frac{1}{3} & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{3}{4} & -\frac{3}{4} \end{array} \right]$$

$$\xrightarrow{R_2 - \frac{5}{3}R_3 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{4} & \frac{5}{4} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{3}{4} & -\frac{3}{4} \end{array} \right] \rightarrow A^{-1} = \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} & \frac{5}{4} \\ -\frac{1}{2} & \frac{3}{4} & -\frac{3}{4} \end{bmatrix}$$

$$[A | I] \xrightarrow{\text{e.r.o}} [I | A^{-1}]$$

2. (34 points) A square matrix A is nilpotent if A^k is the zero matrix for some positive integer k .

If A is nilpotent, what are the possible values of $\det A$?

$$A^k = 0 \rightarrow |A^k| = |0| \rightarrow |A|^k = |0| \rightarrow |A| = 0$$

3. (36 points) Let $A = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 7 & 6 \\ -1 & 1 & -2 & 0 \end{bmatrix}$

(a.) Find a basis for $\text{row}(A)$ and find $\text{rowrank}(A)$.

(b.) Find a basis for $\text{Col}(A)$ and find $\text{colrank}(A)$.

(c.) Find $\text{Null}(A)$ and find $\text{nullity}(A)$.

(d.) Find solutions of $Ax = b$ if $b = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$.

a) $\begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 7 & 6 \\ -1 & 1 & -2 & 0 \end{bmatrix} \xrightarrow{R_1+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 7 & 6 \\ 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{-2R_1+R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{bmatrix}$

$\xrightarrow{-R_2+R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \text{Linearly independent rows} \rightarrow \text{base rows}$

$\rightarrow \text{row}(A) = \text{Span}\{[1 \ 0 \ 3 \ 2], [0 \ 1 \ 1 \ 2]\}$

$\rightarrow \text{row rank}(A) = \underline{2}$

b) The leading one is the first non zero entry in a row $\rightarrow$ The columns containing leading ones are the pivot columns from the original matrix

$\text{col}(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} \quad \text{colrank}(A) = \underline{2}$

c) In order to find $\text{Null}(A)$ we should solve $\begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 7 & 6 \\ -1 & 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\begin{cases} x_1 + 3x_3 + 2x_4 = 0 \\ 2x_1 + x_2 + 7x_3 + 6x_4 = 0 \\ -x_1 + x_2 - 2x_3 = 0 \end{cases} \rightarrow \text{if we take } \begin{cases} x_3 = t \\ x_4 = s \\ x_1 = -2s - 3t \\ x_2 = -2s - t \end{cases} \rightarrow \bar{x} = \begin{bmatrix} -2s - 3t \\ -2s - t \\ t \\ s \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -3 \\ -1 \\ 0 \\ 1 \end{bmatrix} t$

Nullity = 2

d) $Ax=b$, $b = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 7 & 6 \\ -1 & 1 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

$$\rightarrow \begin{cases} x_1 + 3x_3 + 2x_4 = 1 \\ 2x_1 + x_2 + 7x_3 + 6x_4 = 3 \\ -x_1 + x_2 - 2x_3 = 0 \end{cases} \Rightarrow \begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & b \\ \begin{bmatrix} 1 & 0 & 3 & 2 & 1 \\ 2 & 1 & 7 & 6 & 3 \\ -1 & 1 & -2 & 0 & 0 \end{bmatrix} \end{array} \xrightarrow{R_1 + R_3 \rightarrow R_3}$$

$$\begin{bmatrix} 1 & 0 & 3 & 2 & 1 \\ 2 & 1 & 7 & 6 & 3 \\ 0 & 1 & 1 & 2 & 1 \end{bmatrix} \xrightarrow{-2R_1 + R_2 \rightarrow R_2}$$

$$\begin{bmatrix} 1 & 0 & 3 & 2 & 1 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 & 1 \end{bmatrix} \xrightarrow{-R_2 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 3 & 2 & 1 \\ 0 & 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_4 = s \quad x_3 = t \Rightarrow x_2 = 1 - 2s - t \\ x_1 = 1 - 2s - 3t$$

$$\rightarrow X = \begin{bmatrix} 1 - 2s - 3t \\ 1 - 2s - t \\ t \\ s \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ -2 \\ 0 \\ 1 \end{bmatrix} s + \begin{bmatrix} -3 \\ -1 \\ 1 \\ 0 \end{bmatrix} t$$

$$\hookrightarrow \text{Solution is } \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$